DEVELOPMENT OF A CIVIL ENGINEERING DEPARTMENT WEBSITE USING HTML

Web Programming – Assignment Report

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I. INTRODUCTION

A department website is the first place where students and parents look for information. In this assignment a simple website was created for the **Department of Civil Engineering** using HTML. It has a brown header with the department logo, a yellow menu bar, and six full-width sections: About Us, Academic Programmes, Our Faculty, Facilities, Enquiry Form and Contact Details. Alternate sections have a light background so that the reader can easily see where each section begins. The code was written in VS Code and the result was checked in Microsoft Edge.

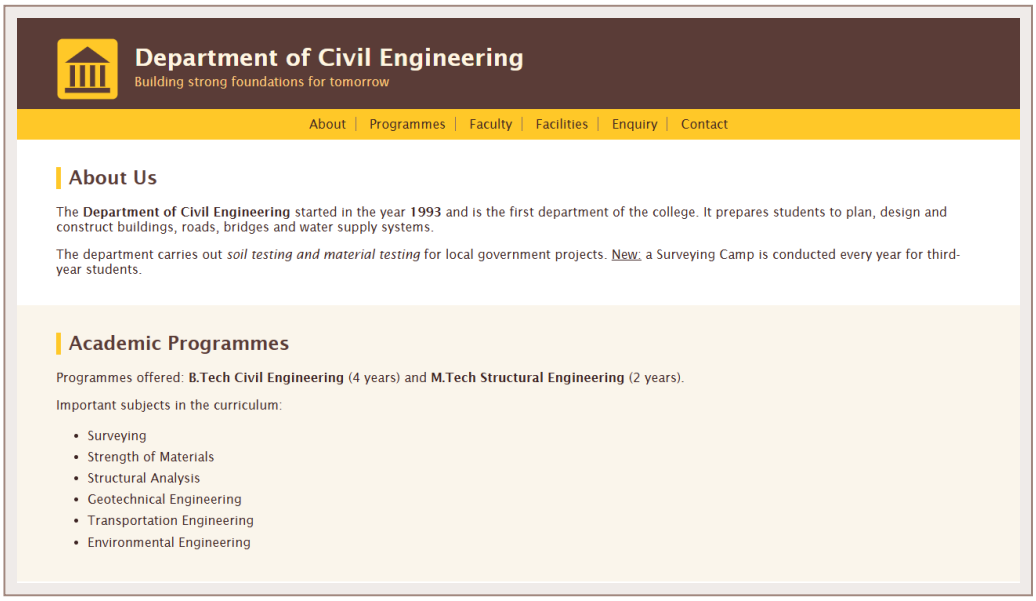


Plate 1. Home page of the Civil Engineering department website

II. HTML DOCUMENT STRUCTURE

An HTML document has a fixed structure. The browser first reads the document type, then the head (information about the page) and finally the body (the content to display).

Listing 1. Skeleton of index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Civil Engineering Department</title>
  <style> ... </style>
</head>
<body>
  ... header, nav, main, footer ...
</body>
</html>
```

<!DOCTYPE html>
Declares the document as HTML5; without it the browser may use "quirks mode".

<html>
The outermost (root) element of the page.

<head>
Holds metadata such as the character set, the title and CSS.

<title>
Sets the name seen on the browser tab.

<body>
Contains all visible elements of the page.

III. HEADINGS, PARAGRAPHS AND TEXT FORMATTING

The page title is written with `<h1>` and the section titles with `<h2>`; HTML supports six levels down to `<h6>`. Text blocks are written in `<p>`. The formatting tags used are `<strong>` (important, bold), `<b>` (bold), `<em>` (emphasis, italic) and `<ins>`, which shows newly inserted text with an underline.

Listing 2. About Us section

```
<h2>About Us</h2>
<p>The <strong>Department of Civil Engineering</strong> started in the year <b>1993</b> ...</p>
<p>The department carries out <em>soil testing and material testing</em> for local government
projects.
   <ins>New:</ins> a Surveying Camp is conducted every year for third-year students.</p>
```

IV. LISTS

Six subjects of the department are listed with an **unordered list**. `<ul>` creates the list and each `<li>` is a bullet point. (An ordered list `<ol>` would number them instead.) The facilities are written with a **description list**: `<dl>`, a term `<dt>` and its description `<dd>`.

Listing 3. List of subjects

```
<ul class="subjects">
  <li>Surveying</li>
  <li>Strength of Materials</li>
  <li>Structural Analysis</li>
  <li>Geotechnical Engineering</li>
  <li>Transportation Engineering</li>
  <li>Environmental Engineering</li>
</ul>
```

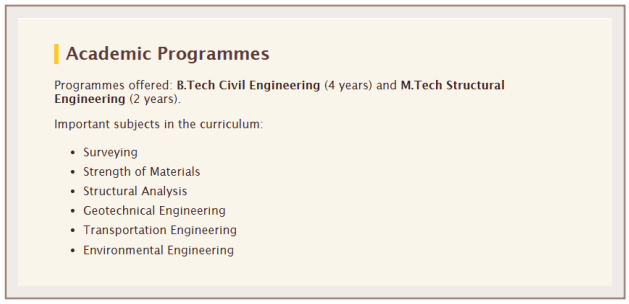


Plate 2. Subjects shown as a bulleted list

V. HYPERLINKS

A hyperlink is created using the anchor tag `<a>`, whose `href` attribute holds the destination. Three types are used: links to sections of the same page (`#faculty`), links to other websites – the college website, the Bureau of Indian Standards and AICTE – opened in a new tab with `target="_blank"`, and a `mailto:` e-mail link.

Listing 4. Navigation and external links

```
<nav>
  <a href="#about">About</a> <a href="#programmes">Programmes</a> <a href="#faculty">Faculty</a>
...
</nav>
See also: <a href="https://www.example.edu" target="_blank">College Website</a>,
<a href="https://www.bis.gov.in" target="_blank">Bureau of Indian Standards</a>,
<a href="https://www.aicte-india.org" target="_blank">AICTE</a>
Email: <a href="mailto:civil@example.edu">civil@example.edu</a>
```

VI. IMAGES

The department logo, showing a building with pillars, is inserted at the left of the header with the `<img>` tag. `src` specifies the image file, `alt` provides a text alternative for visually impaired users and for broken images, and `width` and `height` fix its size at 80 × 80 pixels. The image is moved to the left with the CSS property `float: left`.

Listing 5. Department logo

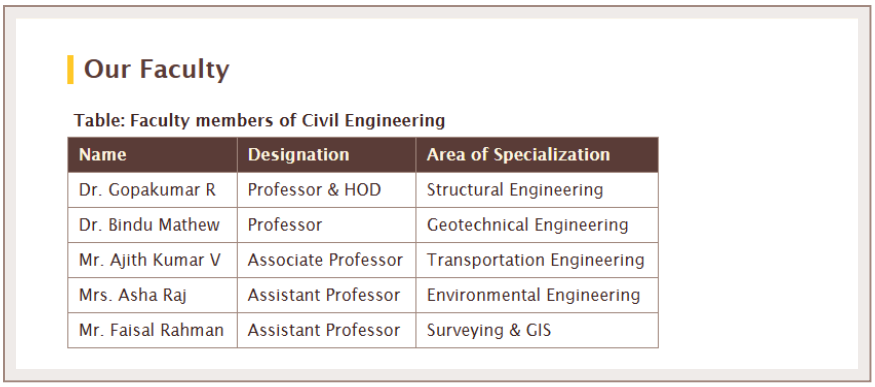
```
<img src="images/civil-logo.svg" alt="Civil Engineering Department Logo" width="80" height="80">
```

VII. TABLES

The faculty table lists five teachers. Besides the basic tags (`<table>`, `<tr>`, `<th>`, `<td>`), it uses `<caption>` for a title and `<thead>`/`<tbody>` to separate the heading row from the data rows.

Listing 6. Faculty table

```
<table>
  <caption>Table: Faculty members of Civil Engineering</caption>
  <thead>
    <tr><th>Name</th><th>Designation</th><th>Area of Specialization</th></tr>
  </thead>
  <tbody>
    <tr><td>Dr. Gopakumar R</td><td>Professor &amp;amp; HOD</td><td>Structural Engineering</td></tr>
    <tr><td>Dr. Bindu Mathew</td><td>Professor</td><td>Geotechnical Engineering</td></tr>
    ... three more rows ...
  </tbody>
</table>
```



Our Faculty

Table: Faculty members of Civil Engineering

Name	Designation	Area of Specialization
Dr. Gopakumar R	Professor & HOD	Structural Engineering
Dr. Bindu Mathew	Professor	Geotechnical Engineering
Mr. Ajith Kumar V	Associate Professor	Transportation Engineering
Mrs. Asha Raj	Assistant Professor	Environmental Engineering
Mr. Faisal Rahman	Assistant Professor	Surveying & GIS

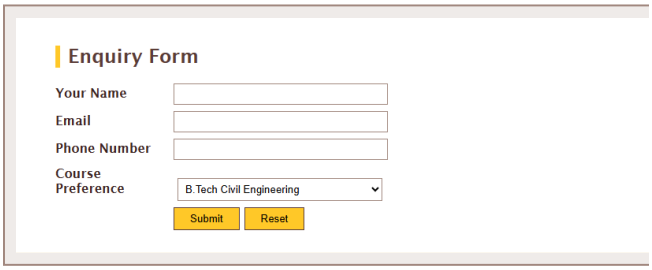
Plate 3. Faculty table as displayed in the browser

VIII. FORMS

The enquiry form is built with `<form action="#" method="post">`. Each field has a `<label>` joined to it by `for`/`id`. The name, e-mail and phone fields use `type="text"`, `type="email"` and `type="tel"`; the phone field has `pattern="[0-9]{10}"` so only a 10-digit number is accepted. The course preference is a `<select>` list whose options are grouped into Undergraduate and Postgraduate using `<optgroup>`. Finally, **Submit** sends the data and **Reset** clears all the fields.

Listing 7. Student enquiry form

```
<form action="#" method="post">
  <p><label for="sn">Your Name</label>
     <input type="text" id="sn" name="name"
required></p>
  <p><label for="se">Email</label>
     <input type="email" id="se" name="email"
required></p>
  <p><label for="sp">Phone Number</label>
     <input type="tel" id="sp" name="phone"
pattern="[0-9]{10}"></p>
  <p><label for="sc">Course Preference</label>
     <select id="sc" name="course">
       <optgroup label="Undergraduate">
         <option>B.Tech Civil
Engineering</option>
       </optgroup>
       <optgroup label="Postgraduate">
         <option>M.Tech Structural
Engineering</option>
       </optgroup>
     </select></p>
  <p><input type="submit" value="Submit">
     <input type="reset" value="Reset"></p>
</form>
```



Enquiry Form

Your Name

Email

Phone Number

Course Preference

B.Tech Civil Engineering

Submit Reset

Plate 4. Enquiry form

IX. SEMANTIC HTML

Semantic elements are tags whose names explain their purpose. They replace a page full of anonymous `<div>` tags with meaningful blocks, which helps accessibility tools, improves search-engine results and makes the code easier to maintain.

Element	Purpose	Used for
`<header>`	Introductory content at the top of a page	Logo, department name and slogan
`<nav>`	A set of navigation links	Yellow menu bar
`<main>`	The unique main content of the page	All six sections
`<section>`	A thematic part of the content with its own heading	About, Programmes, Faculty, Facilities, Enquiry, Contact
`<footer>`	Closing content of the page	Copyright, author and “Back to top” link

X. CONCLUSION

The Civil Engineering department website was successfully created using basic HTML. Each requirement of the assignment – document structure, text formatting, lists, links, images, tables, forms and semantic elements – was used in a practical way, and the finished page is simple, organised and easy to navigate.

REFERENCES

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